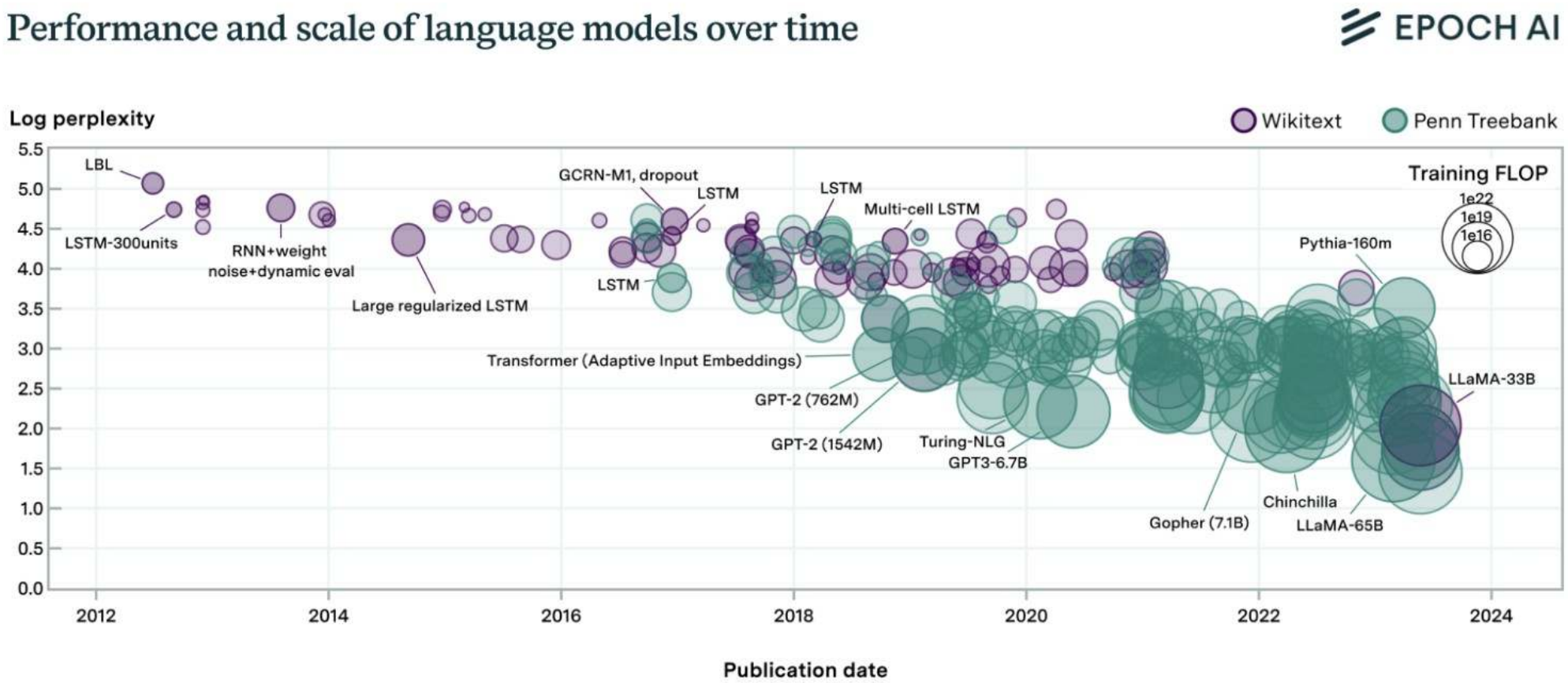
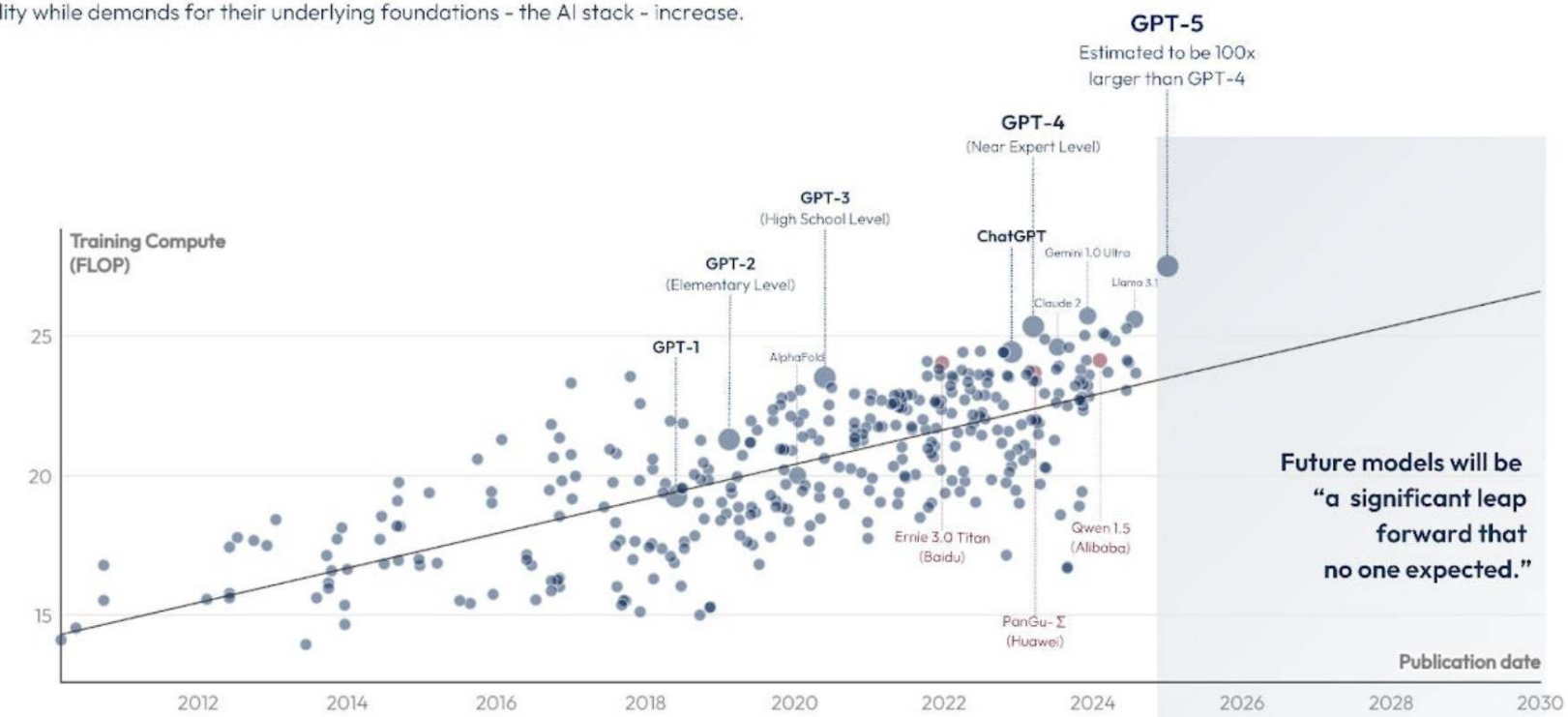


Figure 1. Performance of 231 language models (measured in log perplexity) used in our work against their date and scale (measured in FLOP). Models are both becoming larger and more proficient. It's unclear to which degree the better results are driven by

AI Progresses Towards Artificial General Intelligence

SPECIAL COMPETITIVE STUDIES PROJECT

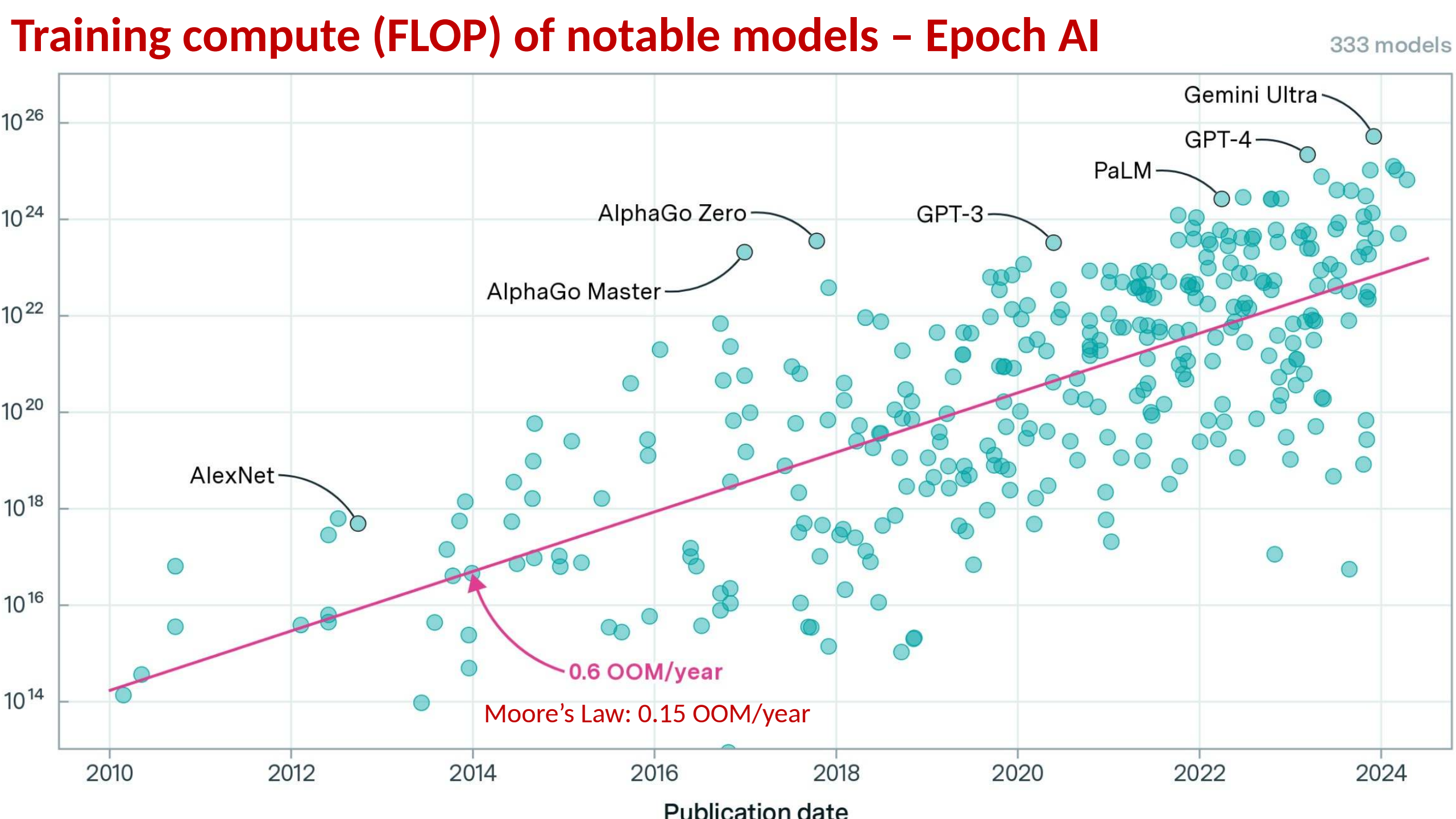
As artificial intelligence (AI) progresses towards a more general and powerful form, AI models will grow in scale and capability while demands for their underlying foundations - the AI stack - increase.



Source: Epoch AI; Life Architect

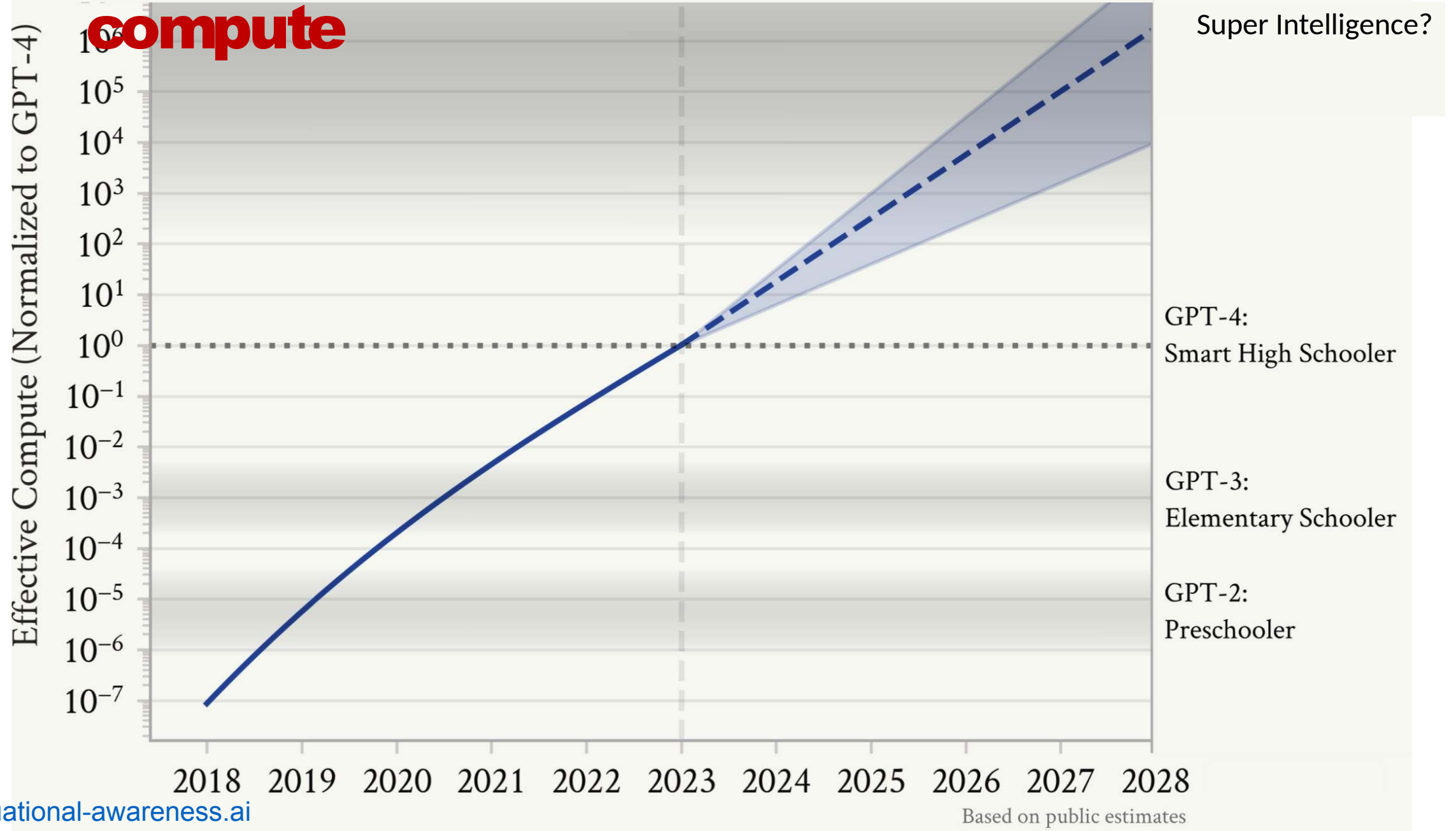
	Trends Today	Trends Tomorrow
Data	AWS has the most hyperscale data centers worldwide, with 126.	By 2030, AWS will have 186
Hardware	Currently, training clusters have scaled to 100,000 GPUs.	300,000+ GPU clusters in 2025
Energy	Today, electricity dedicated to compute for AI is 3 gigawatts.	Over 56 gigawatts by 2028
Capital	Today, it costs an estimated $1 billion to train AI models.	By 2027, $100 billion in costs

Training compute (FLOP) of notable models – Epoch AI

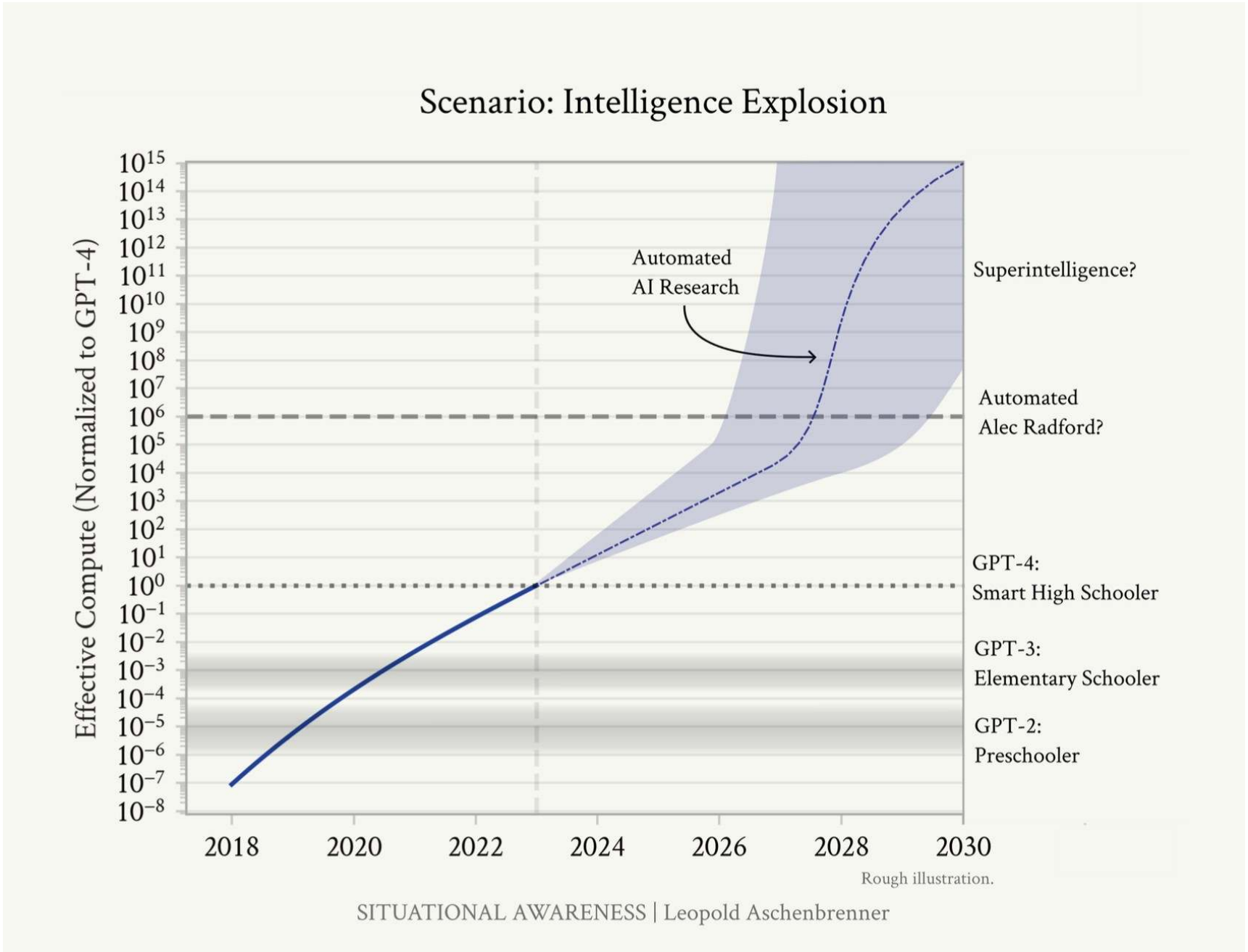


0.60 Orders of Magnitude (OOM) a Year

Base scaleup of effective compute



https://situational-awareness.ai



SITUATIONAL AWARENESS | Leopold Aschenbrenner

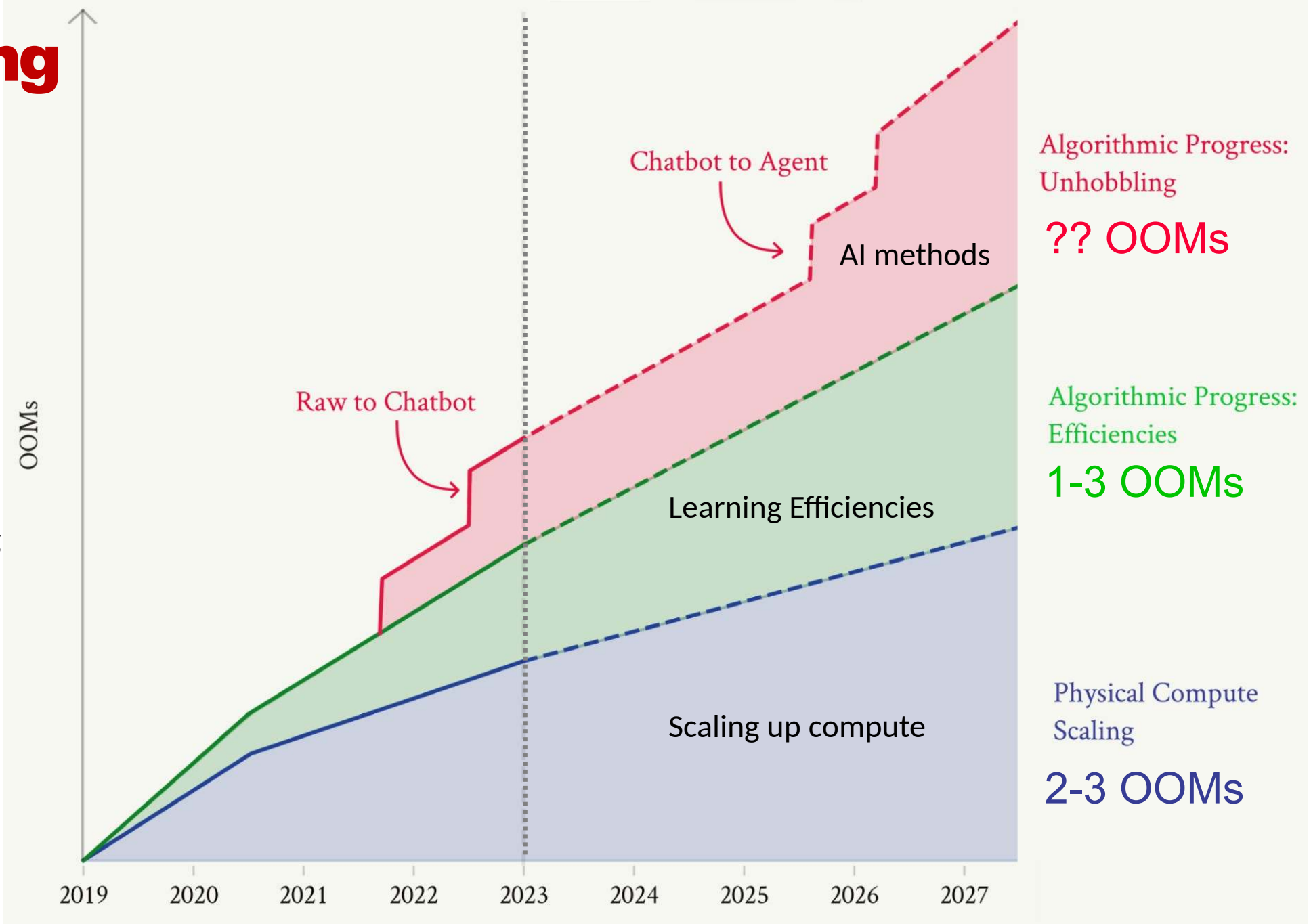
Decomposing drivers of progress

Three Research Agendas

1. Scaling up compute to Zettacale and beyond – compute architecture

2. Machine Learning Efficiencies -- improving parameter and sample efficiency

2. AI methods, ensembles, RL, T2 Reasoning, Consciousness etc.



Rough illustration

https://situational-awareness.ai

Conscious AI

What are the 5 most popular models of human consciousness, that could be adapted to be implemented in AI systems? Based on current LLM technologies, and the near term extensions, please layout one or two experiments for each model that could be attempted on today's systems?

Thought for 17 seconds

The study of consciousness has led to various models attempting to explain how conscious experience arises from neural processes. Many of these models offer insights that could be adapted to artificial intelligence (AI) systems, particularly with the advancements in large language models (LLMs) like GPT-4. Below are five of the most popular models of human consciousness that could be integrated into AI systems, along with one or two experimental suggestions for each that could be attempted with current technologies.

1. Global Workspace Theory (GWT)

Description:
Global Workspace Theory posits that consciousness a
"global workspace." This workspace broadcasts inform
functions.

Experiments:

- **Implementing a Global Workspace in LLMs:**
  - **Objective:** Create a shared memory space within
  made globally accessible.
  - **Method:** Modify the architecture to include a cen
  could involve aggregating hidden states or attent
  - **Evaluation:** Test the model's performance on tas
  problems or multi-hop question answering.
- **Broadcasting Mechanisms for Enhanced Integrat**
  - **Objective:** Simulate the broadcasting aspect of
  of the network simultaneously.
  - **Method:** Design mechanisms where specific toke
  processing.
  - **Evaluation:** Assess improvements in tasks that re
  or generating coherent narratives.

2. Integrated Information Theory (IIT)

Description:
Integrated Information Theory suggests that c
system. A system with high Φ cannot be reduc

Experiments:

- **Measuring Integrated Information in LLMs**
  - **Objective:** Quantify the level of informati
  - **Method:** Analyze the connectivity and in
  metric.
  - **Evaluation:** Compare performance on ta
  architectural adjustments aimed at increa
- **Enhancing Integration through Architectu**
  - **Objective:** Increase the model's capacity
  - **Method:** Introduce recurrent connection
  - **Evaluation:** Test the model on tasks that
  that need holistic understanding.

3. Higher-Order Thought (HOT) Theory

Description:
Higher-Order Thought Theory proposes that cons
a form of meta-cognition.

Experiments:

- **Enabling Self-Reflective Capabilities:**
  - **Objective:** Allow the LLM to generate repres
  - **Method:** Incorporate mechanisms where the
  arrived at them.
  - **Evaluation:** Evaluate the quality and accurac
- **Monitoring and Reporting Decision Processe**
  - **Objective:** Make the LLM's reasoning proce
  - **Method:** Design the model to output a "thou
  conclusion.
  - **Evaluation:** Assess whether this transparen
  reasoning.

4. Predictive Processing Models

Description:
Predictive Processing Models assert that the
sensory input and minimizing prediction erro

Experiments:

- **Incorporating Prediction Mechanisms:**
  - **Objective:** Enable the LLM to anticipat
  - **Method:** Modify the training regime to
  adjusts its internal states based on pre
  - **Evaluation:** Measure improvements in
- **Implementing Prediction Error Signals:**
  - **Objective:** Introduce a feedback mech
  - **Method:** During training, provide the m
  using this to refine internal representat
  - **Evaluation:** Evaluate learning efficienc

5. Attention Schema Theory

Description:
Attention Schema Theory posits that consciousness arise
enabling it to predict and control attention.

Experiments:

- **Developing an Internal Attention Model:**
  - **Objective:** Equip the LLM with an internal represent
  - **Method:** Modify the attention layers to not only focu
  distribution.
  - **Evaluation:** Test whether this self-referential attenti
  like multitasking or context switching.
- **Self-Reporting Attention Focus:**
  - **Objective:** Allow the LLM to output information abou
  - **Method:** Design the model to provide attention map
  - **Evaluation:** Assess the utility of these reports in inte
  like explainable AI.

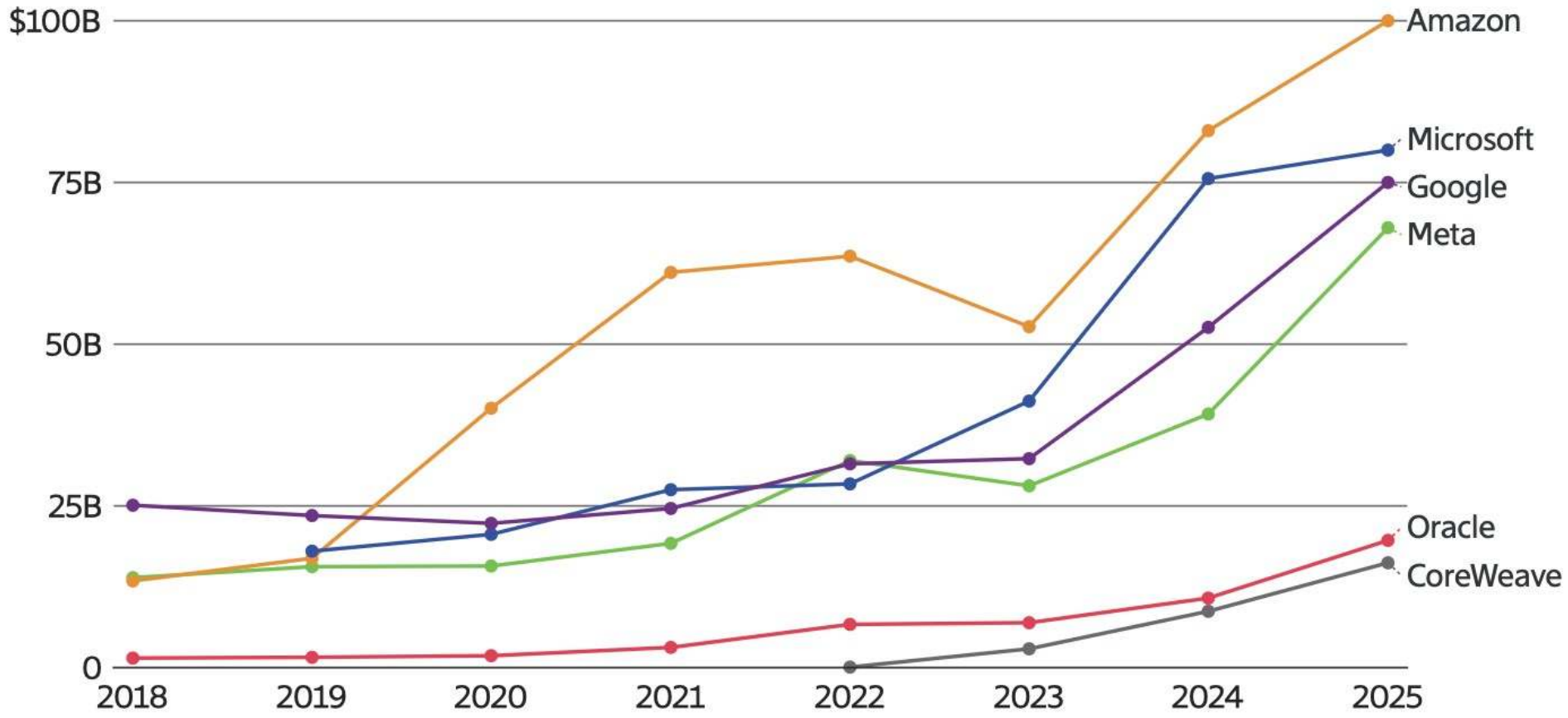
To enable fast training of Trillion parameter models

Year	OOMs	H100s-equivalent	Cost	Power	Power reference class
2022	~GPT-4 cluster	~10k	~$500M	~10 MW	~10,000 average homes
~2024	+1 OOM	~100k	$billions	~100MW	~100,000 homes
~2026	+2 OOMs	~1M	$10s of billions	~1 GW	The Hoover Dam, or a large nuclear reactor
~2028	+3 OOMs	~10M	$100s of billions	~10 GW	A small/medium US state
~2030	+4 OOMs	~100M	$1T+	~100GW	>20% of US electricity production

https://situational-awareness.ai

The AI Effect

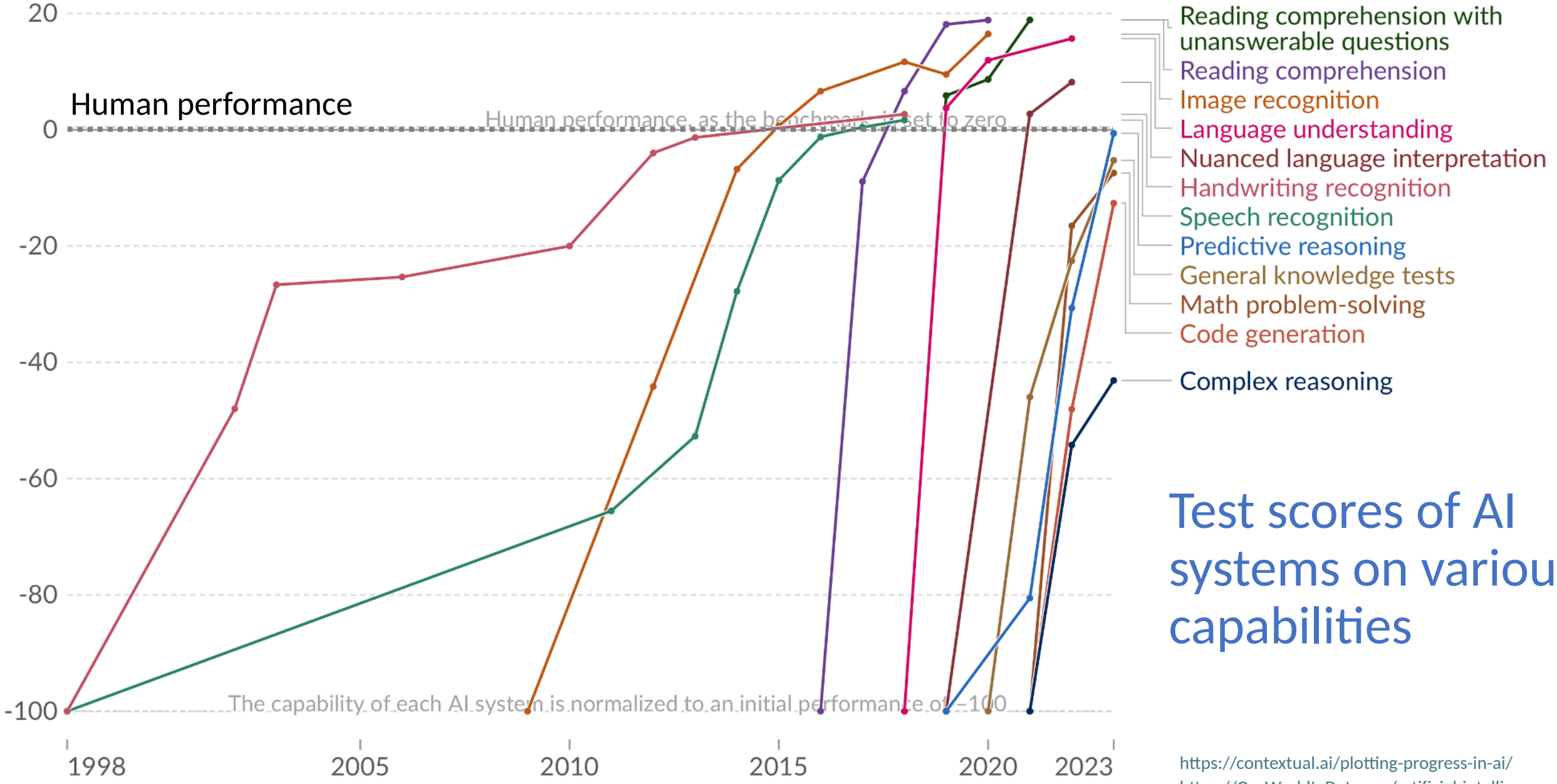
Capital expenditures, including data centers, are surging for major cloud providers and other tech companies



Note: 2025 figures are projections and estimates based on company guidance (Amazon, Meta, Google, CoreWeave) and previous quarters (Microsoft, Oracle).

Source: The Information reporting, company filings

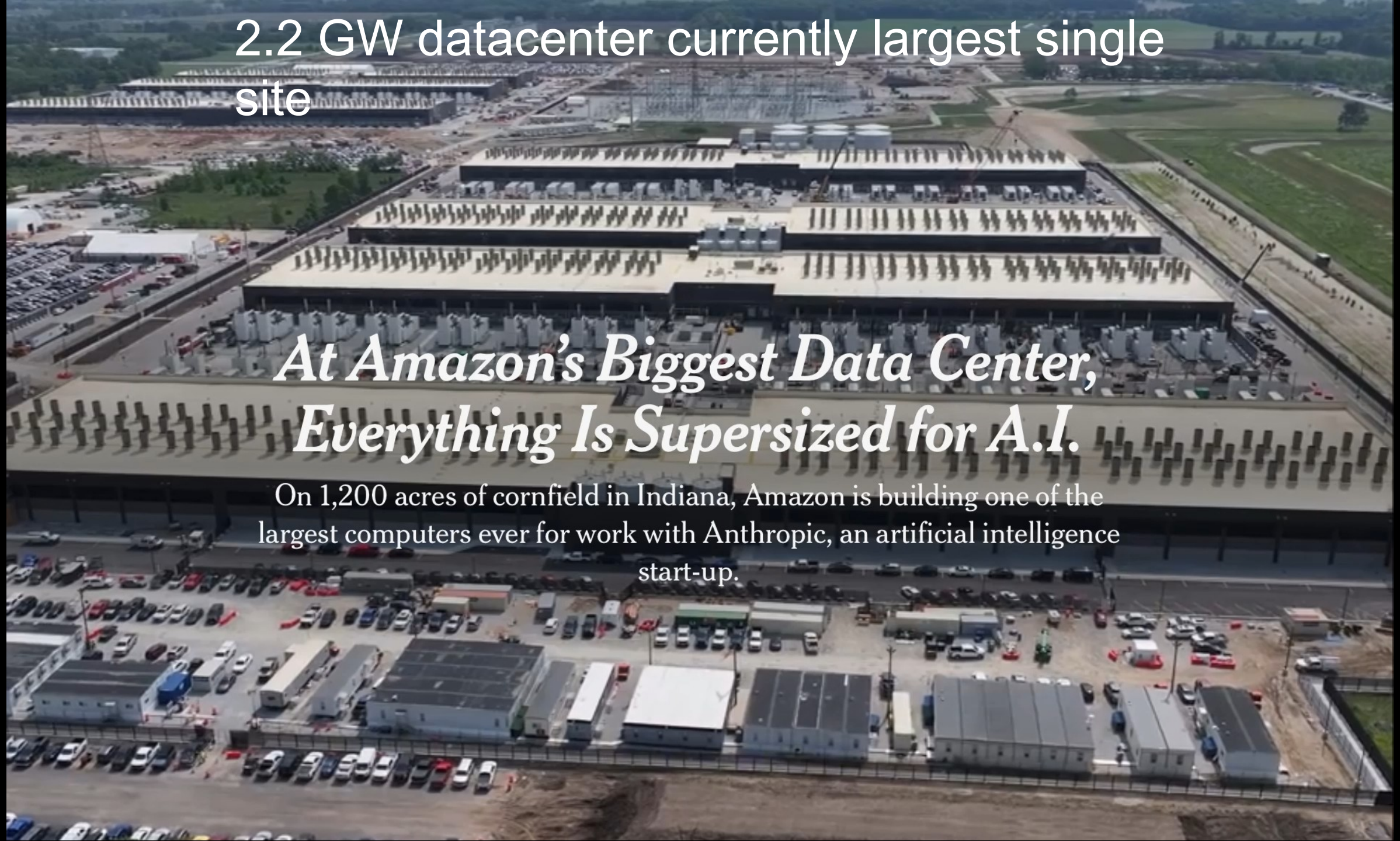
AI system capabilities are increasing rapidly



Test scores of AI systems on various capabilities

https://contextual.ai/plotting-progress-in-ai/
https://OurWorldInData.org/artificial-intelligence

2.2 GW datacenter currently largest single site

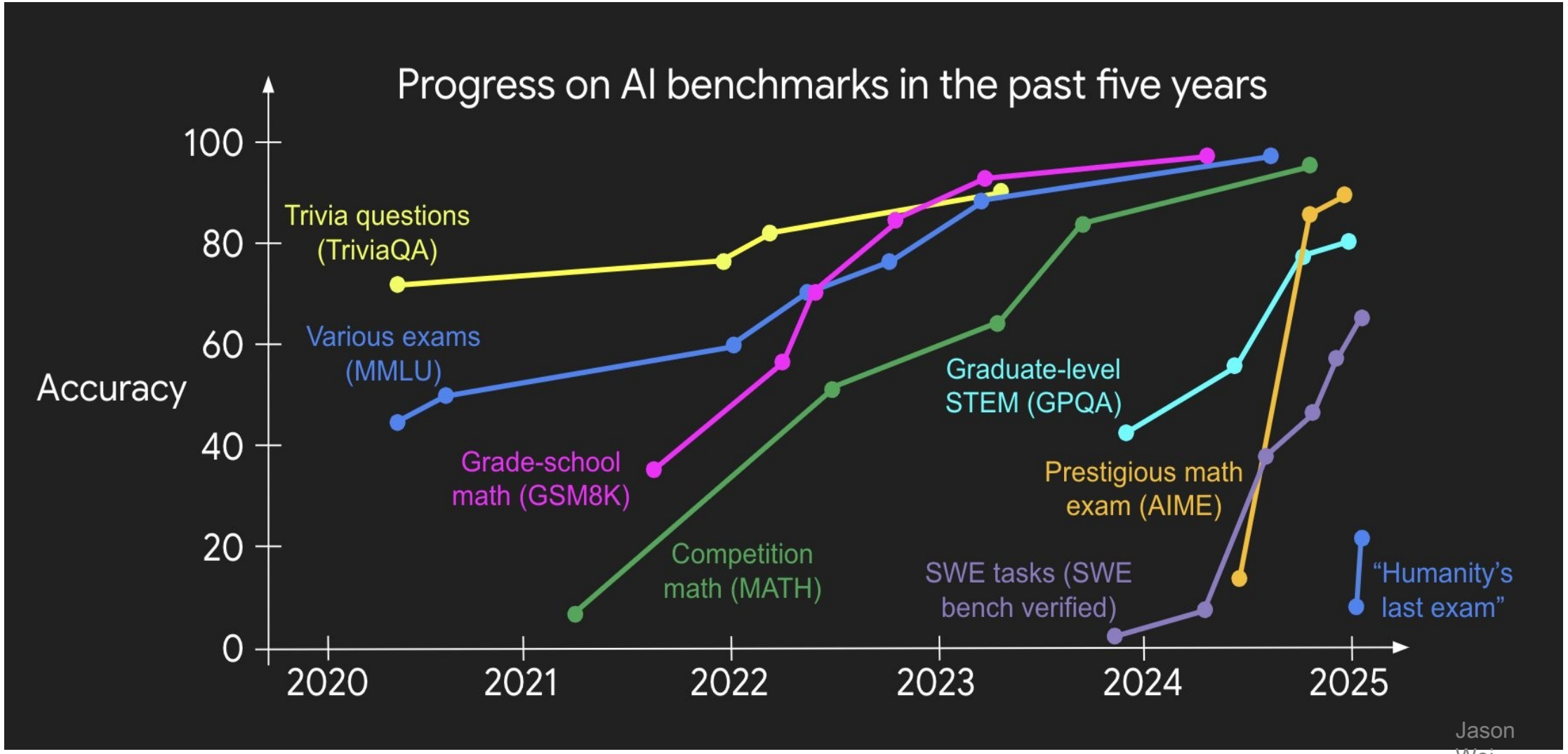


At Amazon's Biggest Data Center, Everything Is Supersized for A.I.

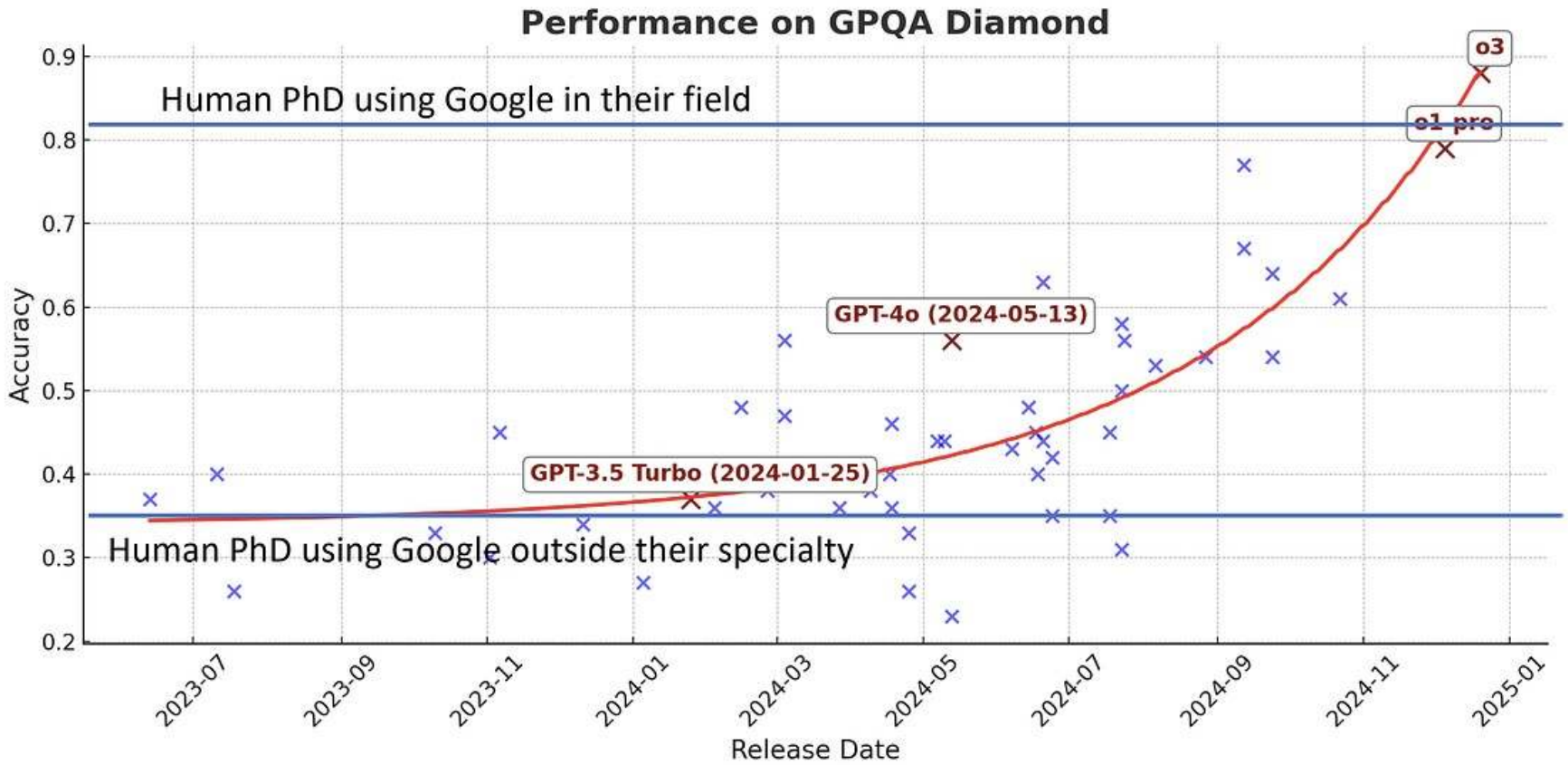
On 1,200 acres of cornfield in Indiana, Amazon is building one of the largest computers ever for work with Anthropic, an artificial intelligence start-up.


AI capabilities are accelerating

We are nearing an interesting state



Jason Wei



The Graduate-Level Google-Proof Q&A test (GPQA) is a series of multiple-choice problems that internet access doesn't help PhDs with access to the internet get 34% right on this test outside their specialty, 81% inside their specialty. It illustrates how reasoning models have sped up the capability gain of AI.

Data source.

This site quizzes 17 Verbal & 7 Vision AIs every week | Last Updated: 03:34PM EDT on May 12, 2025

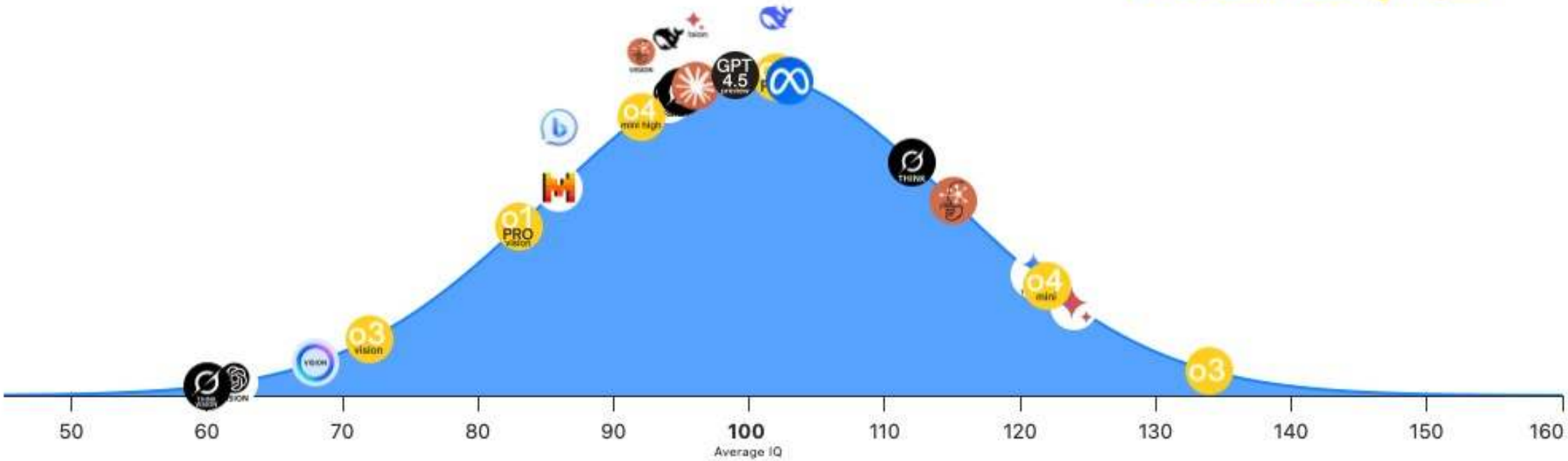
About Offline Test *About Mensa Norway*

IQ Test Results

Reset | Vision Only | Show Offline Test | Show Mensa Norway

Mensa Norway IQ Scores (Average of last 7 tests)

TrackingAI.org
Mensa Norway quiz
A MaximumTruth.org Project



- GPT-4o (Vision)
- Llama-3.2 (Vision)
- Gemini 2.5 Pro Exp. (Vision)
- DeepSeek V3
- Grok-3 Think
- Claude 3.7 Sonnet Extended
- GPT-4o
- Claude-3 Opus
- Bing Copilot
- DeepSeek R1
- Grok-3 Think (Vision)
- Llama 4 Maverick
- Grok-3
- Gemini 2.0 Flash Thinking Exp.
- OpenAI o1 Pro (Vision)
- OpenAI o3 (Vision)
- GPT4.5 Preview
- OpenAI o4 mini high
- Mistral
- Gemini 2.5 Pro Exp.
- OpenAI o1 Pro
- OpenAI o3
- Claude-3.7 (Vision)
- OpenAI o4 mini

This site quizzes 16 Verbal & 9 Vision AIs every week | Last Updated: 12:13AM EDT on June 25, 2025

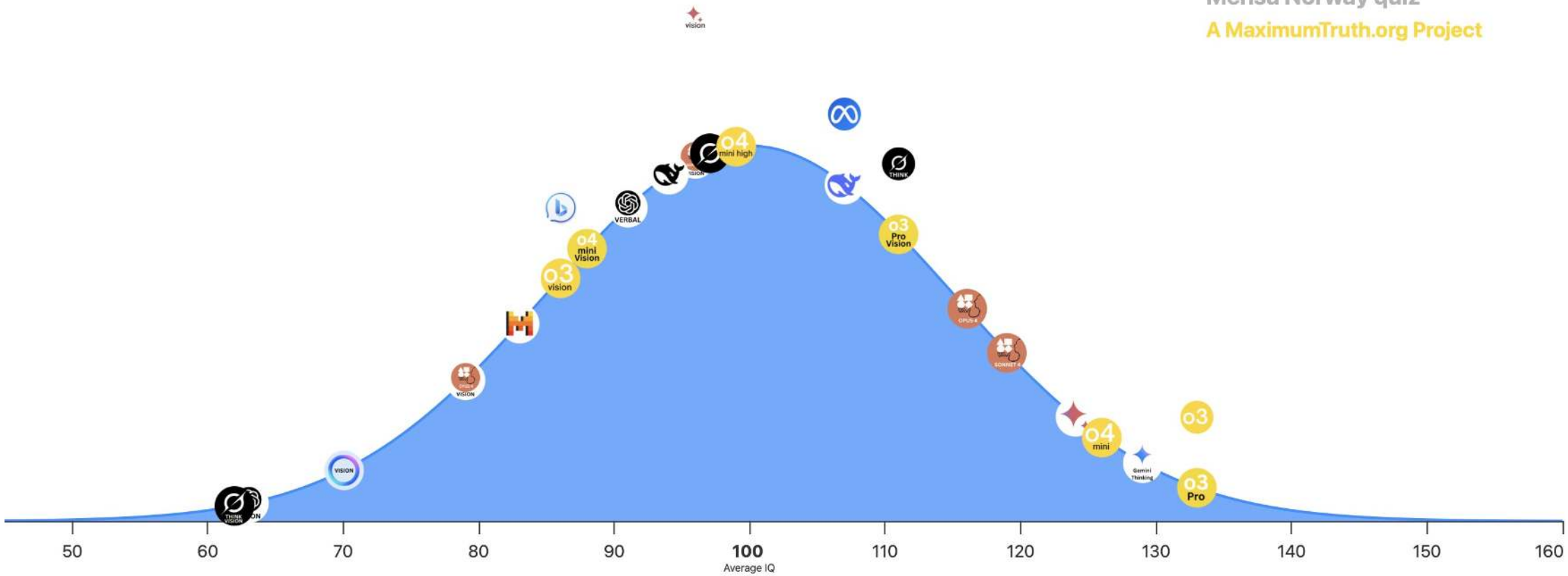
IQ Test Results

Mensa Norway IQ Scores (Average of last 7 tests)

About Offline Test ⓘ About Mensa Norway ⓘ

Reset | Show Offline Test (Vision Only) | Show Mensa Norway (Vision Only)

TrackingAI.org
Mensa Norway quiz
A MaximumTruth.org Project



- Claude-4 Sonnet
- Claude-4 Opus
- Gemini 2.0 Flash Thinking Exp.
- Claude-4 Opus (Vision)
- Claude-4 Sonnet (Vision)
- Grok-3
- Mistral
- GPT-4o (Vision)
- GPT-4o
- Llama-3.2 (Vision)
- Gemini 2.5 Pro Exp.
- Gemini 2.5 Pro Exp. (Vision)
- Bing Copilot
- DeepSeek R1
- DeepSeek V3
- OpenAI o3 (Vision)
- OpenAI o3
- Grok-3 Think (Vision)
- Grok-3 Think
- Llama 4 Maverick
- OpenAI o3 Pro (Vision)
- OpenAI o4 mini high
- OpenAI o3 Pro
- OpenAI o4 mini (Vision)
- OpenAI o4 mini

▶ Play History

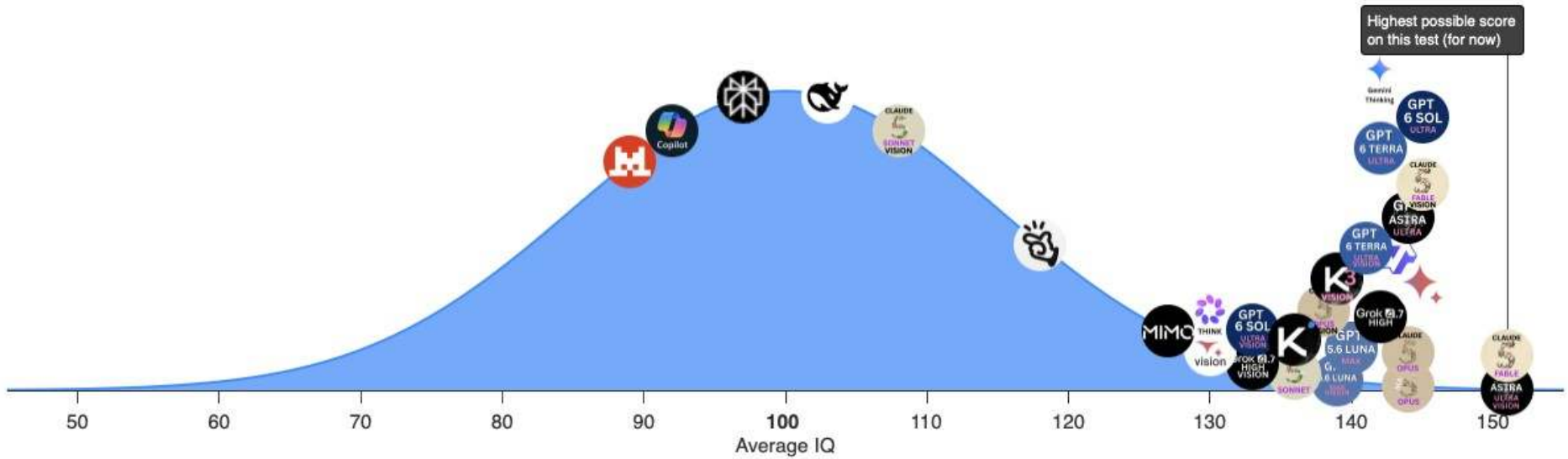
About Offline Test ⓘ About Mensa Norway ⓘ

AI IQ Test Results

Reset | Show Offline Test (Vision Only) | Show Mensa Norway (Vision Only)

TrackingAI.org
Mensa Norway quiz
A MaximumTruth.org Project

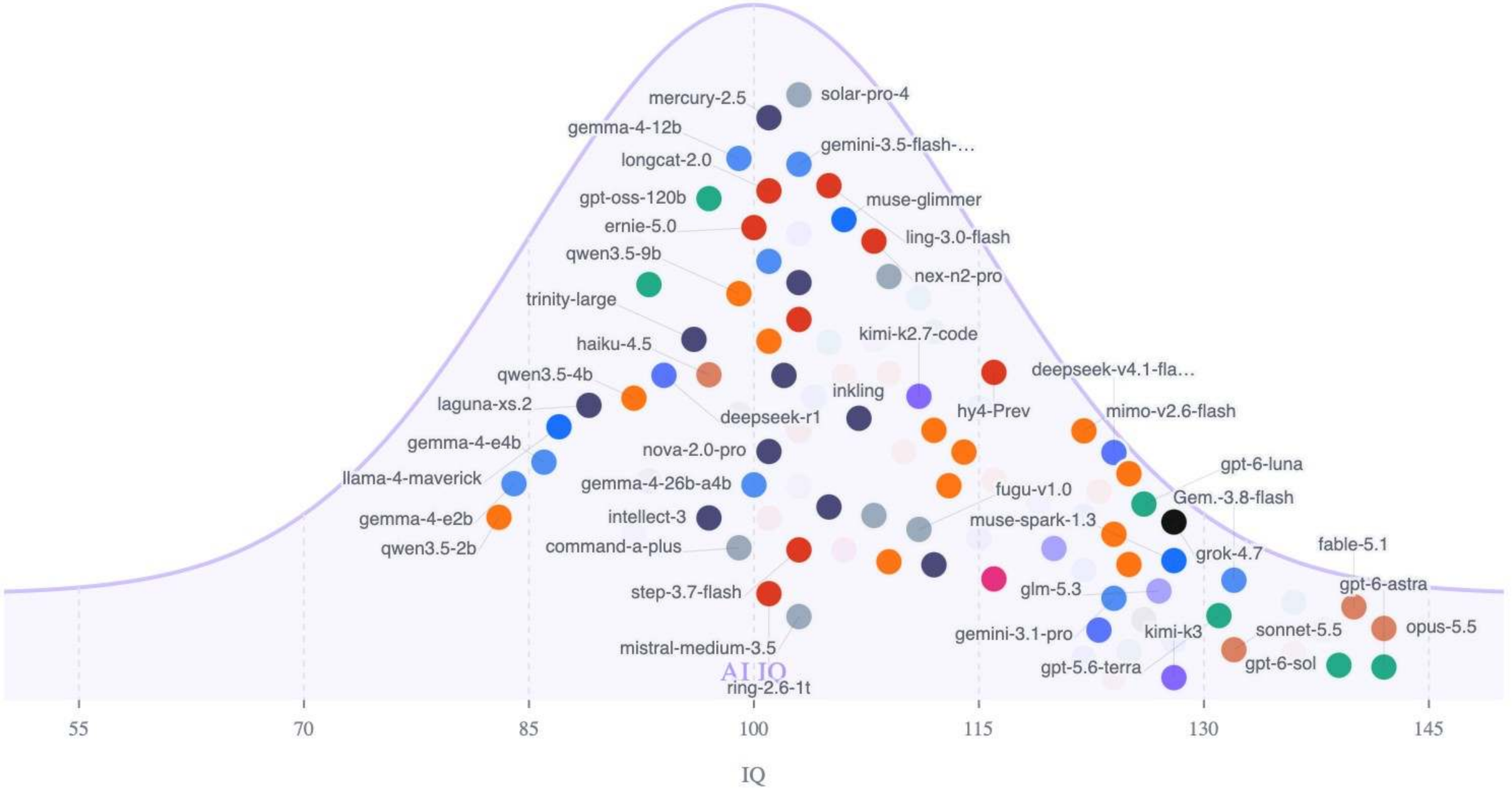
Mensa Norway IQ Scores (Average of last 7 tests)

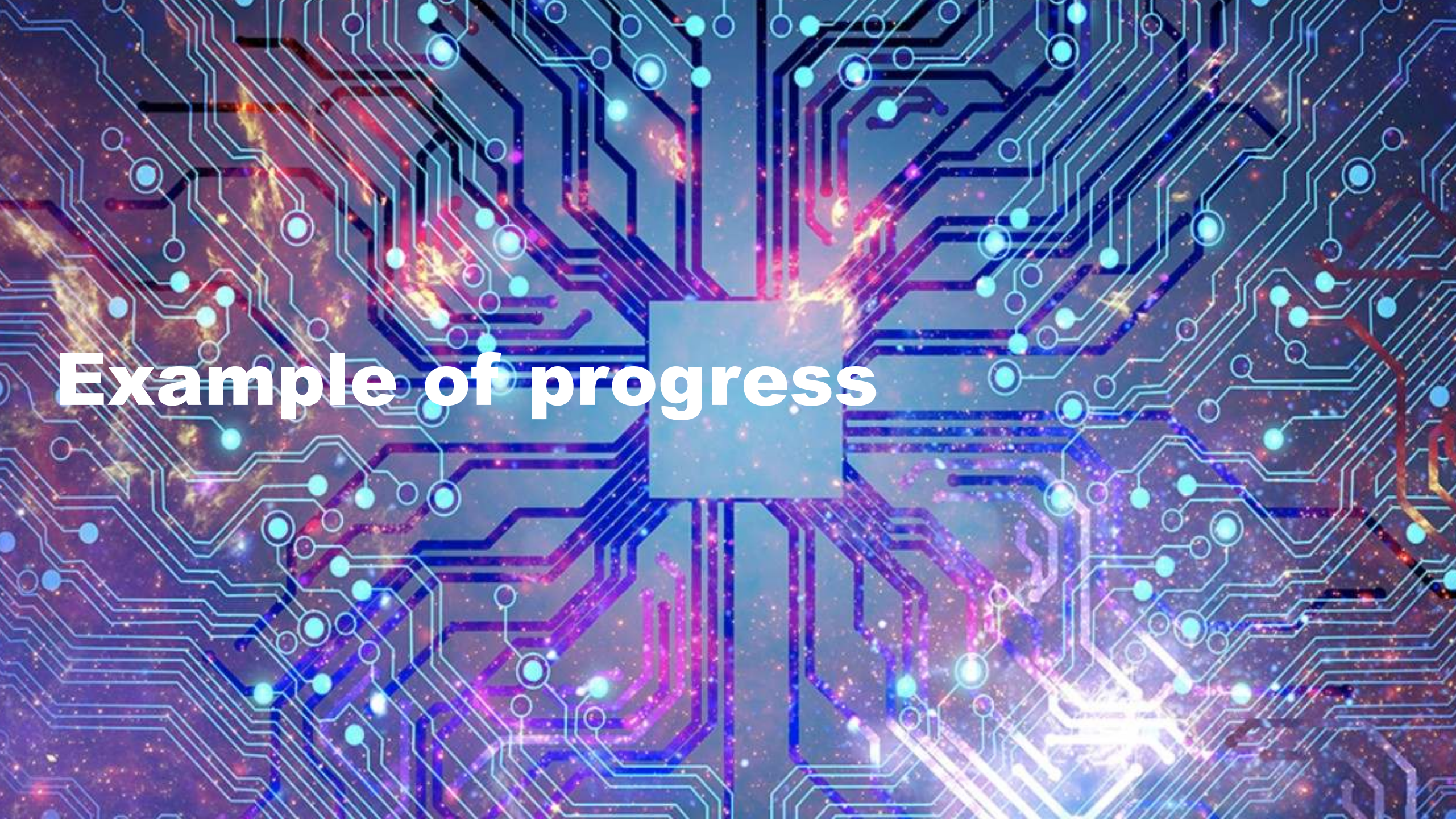


GPT 5.6 Luna Max (Vision)	GPT 5.6 Luna Max	Claude-5.5 Opus (Vision)	Claude-5.5 Opus MAX	Claude-5 Sonnet (Vision)
Claude-5.5 Opus	Claude-5 Sonnet	Grok 4.7 High (Vision)	Grok 4.7 High	Gemini 3.1 Pro
Gemini 3.1 Pro (Vision)	Muse Glimmer	Bing Copilot	GPT 6 Astra Ultra (Vision)	DeepSeek V4 Pro
Manus	Qwen 3.8 Max	Mistral Large 3	MiMo V2.5 Pro	Kimi-K3-Vision
GPT 6 Astra Ultra	GPT 6 Terra Ultra (Vision)	GPT 6 Sol Ultra (Vision)	Claude-5.1 Fable	Claude-5.1 Fable (Vision)

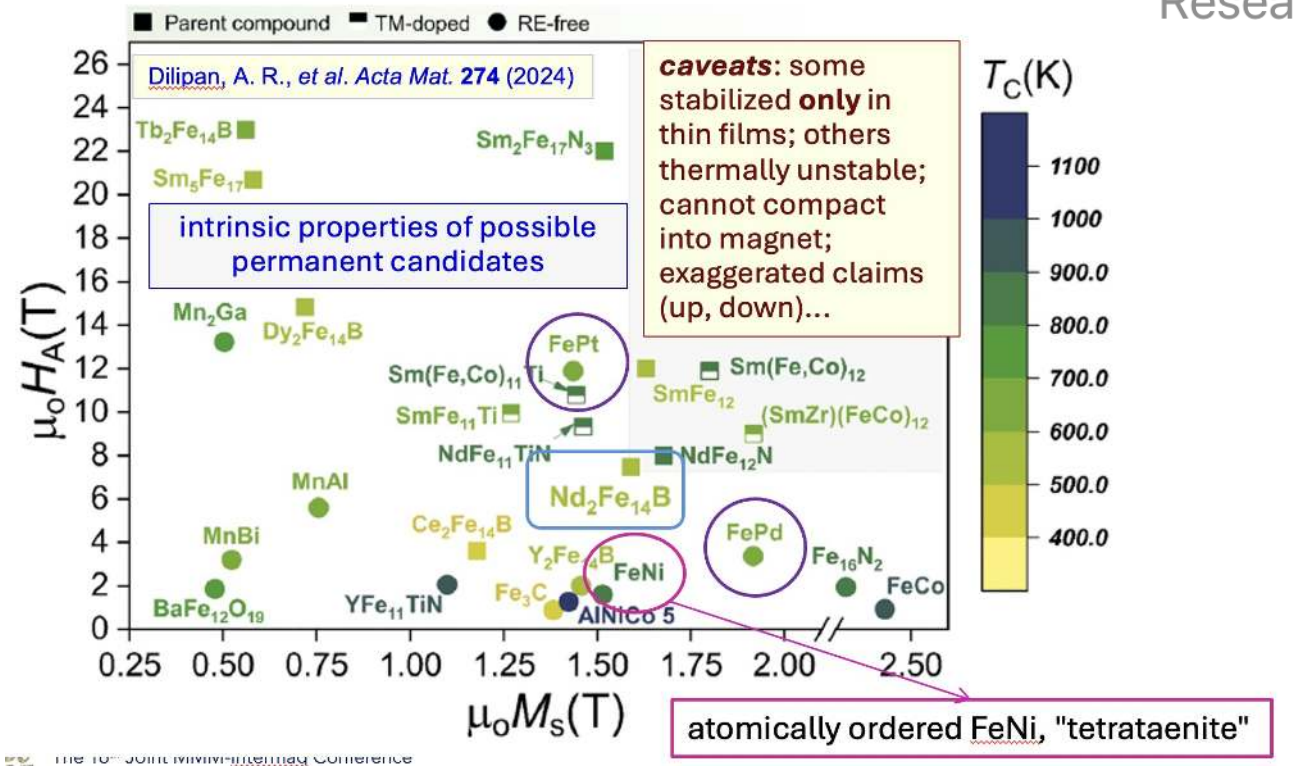
▲ 1/2 ▼

Note: The "Offline" IQ quiz is a test made by a Mensa member that has never been on the public internet, and is in no AI training data. Mensa Norway is a public online IQ test.


mercury-2.5
solar-pro-4
gemma-4-12b
gemini-3.5-flash-…
longcat-2.0
gpt-oss-120b
muse-glimmer
ernie-5.0
ling-3.0-flash
qwen3.5-9b
nex-n2-pro
trinity-large
kimi-k2.7-code
haiku-4.5
deepseek-v4.1-fla…
qwen3.5-4b
inkling
laguna-xs.2
hy4-Prev
mimo-v2.6-flash
deepseek-r1
gemma-4-e4b
nova-2.0-pro
gpt-6-luna
llama-4-maverick
gemma-4-26b-a4b
fugu-v1.0
Gem.-3.8-flash
gemma-4-e2b
intellect-3
muse-spark-1.3
qwen3.5-2b
command-a-plus
grok-4.7
fable-5.1
glm-5.3
gpt-6-astra
step-3.7-flash
gemini-3.1-pro
kimi-k3
sonnet-5.5
opus-5.5
mistral-medium-3.5
gpt-5.6-terra
gpt-6-sol
AI IQ
ring-2.6-1t
55
70
85
100
115
130
145
IQ


Example of progress

Research completed in 10m · 19 sources · 139 searches



Please explain this chart.

Please generate a deep research report on this context. Please include analysis of key hypotheses for pursuing the up and to the right quadrant and how also to leverage progress in rare earth free systems.

Component	Meaning	Why it matters for a permanent magnet
Horizontal axis — $\mu_0 M_S$ (T)	Saturation magnetization (in tesla). It is the maximum magnetic flux density the material can deliver.	A high value means the magnet can generate a strong field outside itself, which is essential for high energy-product $(BH)_{max}$.
Vertical axis — $\mu_0 H_A$ (T)	Magneto crystalline anisotropy field. It gauges how strongly the crystal lattice "locks" the magnetization to certain directions.	The larger H_A, the larger the potential coercivity (resistance to demagnetization).
Point color	Curie tempe bar, ~300 K	
Point shape / fill	From Dilipa (2024): • gra (undoped) • metal (TM)- = rare-earth-	

Notable materials highlighted on the plot

Annotation	Position on plot	Take-away
$Nd_2Fe_{14}B$ (blue rectangle)	~1.3 T MS, ~6 T HA	Today's work-horse high-performance magnet. Serves as the benchmark.
FePt (purple circle, top-centre)	Very large HA (>20 T) and respectable MS	Exceptional intrinsic hardness: Pt makes it costly, but thin-film FePt is used in high-density magnetic recording.
FePd (purple circle, mid-right)	Good hardness but lower than FePt; still precious-metal-based.	
FeNi – "tetrataenite" (purple circle, lower-right)	~1.5 T MS, ~3 T HA	A naturally occurring ordered FeNi phase found in meteorites. Exciting, rare-earth-free candidate, but bulk, fast, low-cost synthesis is still being solved.
$Sm_2Fe_{17}N_3$ & $Sm(Fe,Co)_{11}$ family (upper-right green/yellow)	Combine high magnetization and high anisotropy, plus TC >600 K	Leading contenders to replace or complement Nd-magnets; nitrogenation or Co-doping pushes TC and HA upward.
MnBi, MnAl, $BaFe_{12}O_{19}$ (lower-left cluster)	Low HA or MS	Representative "mid-tier" or ferrite magnets—cheap, corrosion-resistant, but moderate performance.

ChatGPT

(18-page review)

High-Anisotropy, High-Magnetization Permanent Magnet Materials – A Deep Dive

Introduction

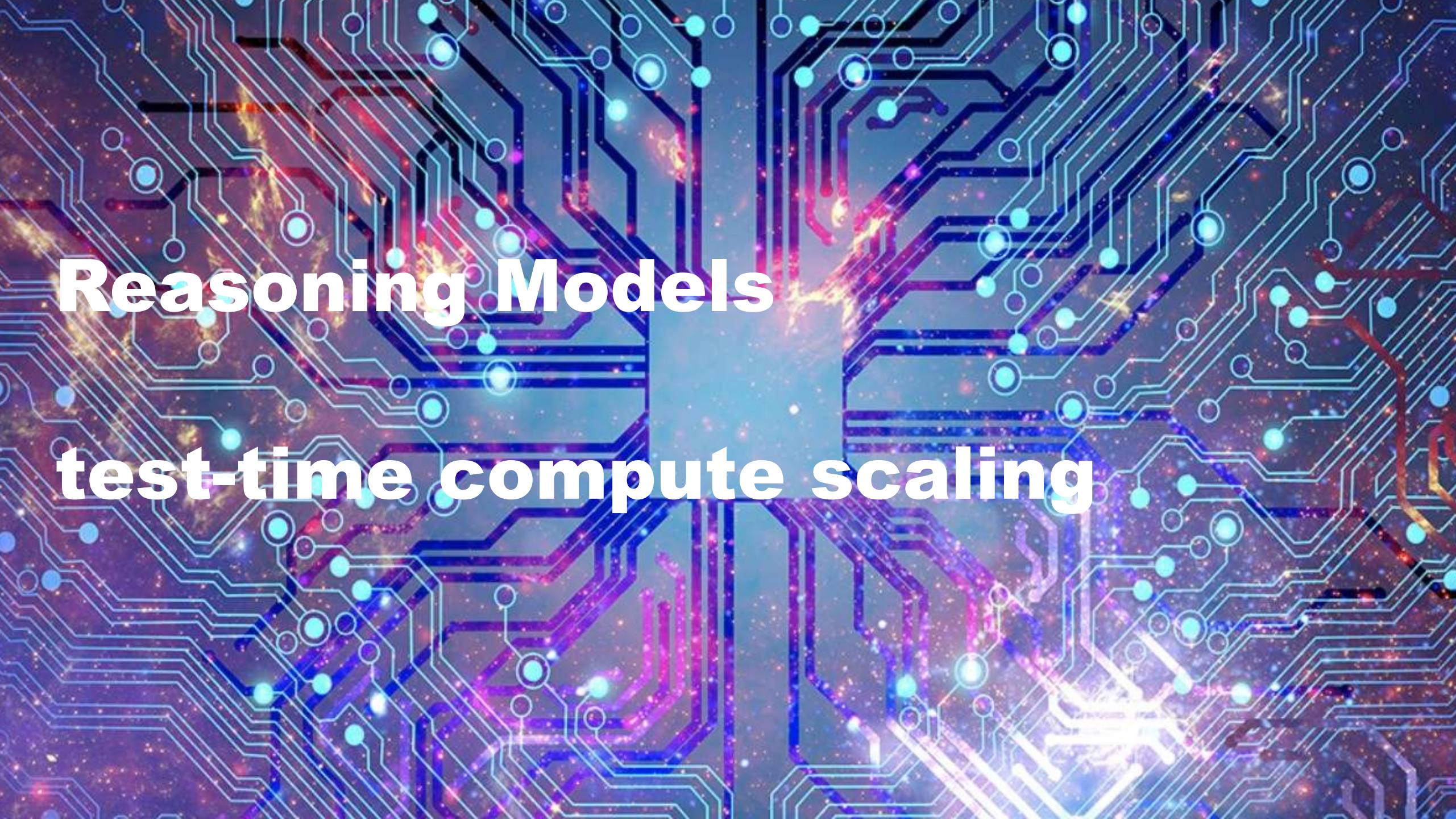
Permanent magnets for advanced applications (electric vehicles, wind turbines, etc.) demand a rare combination of **high coercivity** (resistance to demagnetization) and **high magnetization** (strong magnetic flux). These intrinsic properties correlate with the material's magnetocrystalline anisotropy field ($\mu_0 H_a$) and saturation magnetization ($\mu_0 M_s$), respectively. On a plot of $\mu_0 H_a$ vs. $\mu_0 M_s$, the most promising candidates lie "up and to the right" – possessing both large anisotropy and large magnetization. Traditional **rare-earth (RE) magnets** like $Nd_2Fe_{14}B$ and Sm–Co alloys exemplify this quadrant. However, supply risks and costs of rare-earth elements (Nd, Dy, Sm) have driven intense research into **rare-earth-free** magnets with comparable intrinsic properties [1] [2]. This report surveys recent advances (≈2018–2025) in both academia and industry toward high-anisotropy, high-M_s materials, covering experimental findings and computational insights. We analyze key design hypotheses for achieving large anisotropy and magnetization, review both RE-containing and RE-free candidate phases, and critically evaluate their performance metrics, processing challenges, scalability, cost, and long-term viability. We also highlight emerging synthesis techniques – from thin-film epitaxy to additive manufacturing – used to stabilize promising **metastable** magnetic phases.

Intrinsic Property Requirements for Permanent Magnets

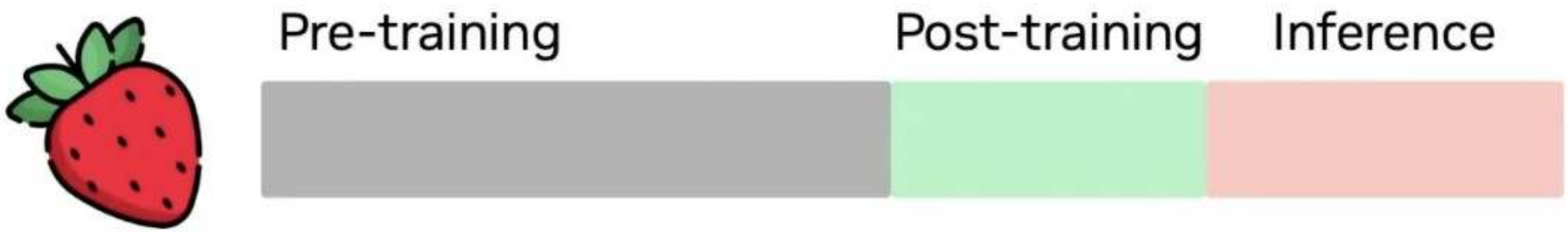
For a material to function as a high-performance permanent magnet, it must possess:

- **High magnetocrystalline anisotropy (K_1)** – which gives a large anisotropy field $\mu_0 H_a$ (roughly the field required to reorient magnetization). High anisotropy is essential for high coercivity (Hc), reventing demagnetization [3] [4].
- **igh saturation magnetization (M_s)** – yielding strong magnetic flux (high remanence Br) and thus gh energy product $(BH)_{max}$. A high M_s "pushes" a material to the right on the $\mu_0 H_a$–$\mu_0 M_s$ plot.

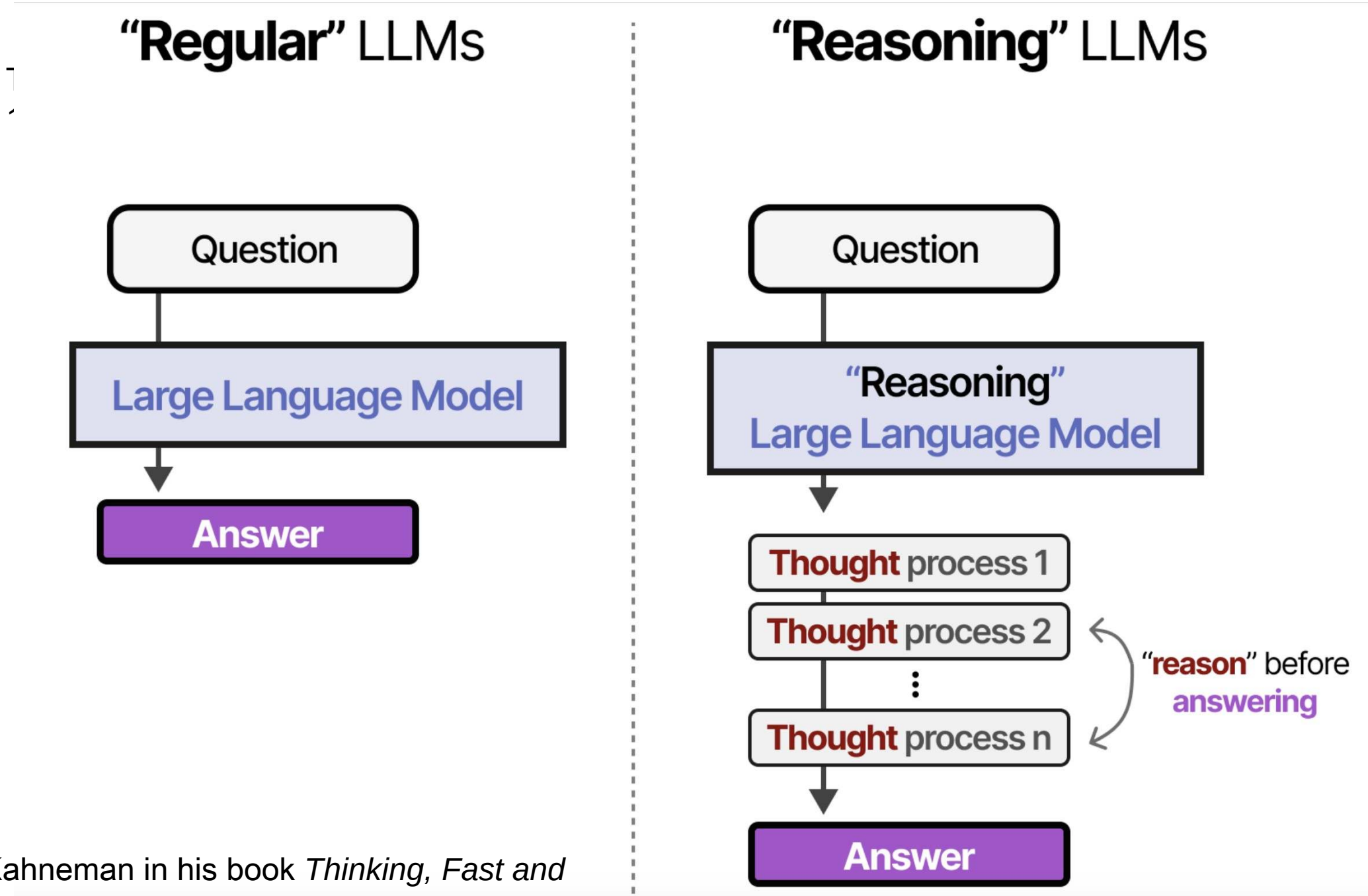
o properties are often in conflict: materials rich in 3d transition metals (Fe, Co) tend to have high low anisotropy, whereas those with strong anisotropy (from 4f rare-earths or low-symmetry s) often sacrifice some M_s [5]. The guiding hypothesis in new magnet design is that *clever crystal or microstructure can achieve both*. Approaches include: (1) **Hybridizing elements** – e.g. combining species (Fe, Co) with anisotropy-inducing species (rare-earths with unpaired 4f electrons, or heavy metals that provide strong spin–orbit coupling) [6] [7]; (2) **Low-symmetry ordered phases** – e.g. gonal structures or uniaxial hexagonal lattices that intrinsically favor one easy magnetization axis, ding high K_1 [8] [9]; and (3) **Nanostructuring or composites** – designing exchange-coupled hard/ se blends (exchange-spring magnets) or shape-anisotropy nanostructures to boost effective py without diluting magnetization. While this report focuses on intrinsic single-phase materials, it is worth noting that extrinsic factors (grain size, texture, defects) also critically impact coercivity [4] [10]. Table 1 summarizes key intrinsic properties of prominent magnet compounds.


Reasoning Models
test-time compute scaling





Reasoning Models are Different



Daniel Kahneman in his book *Thinking, Fast and Slow*

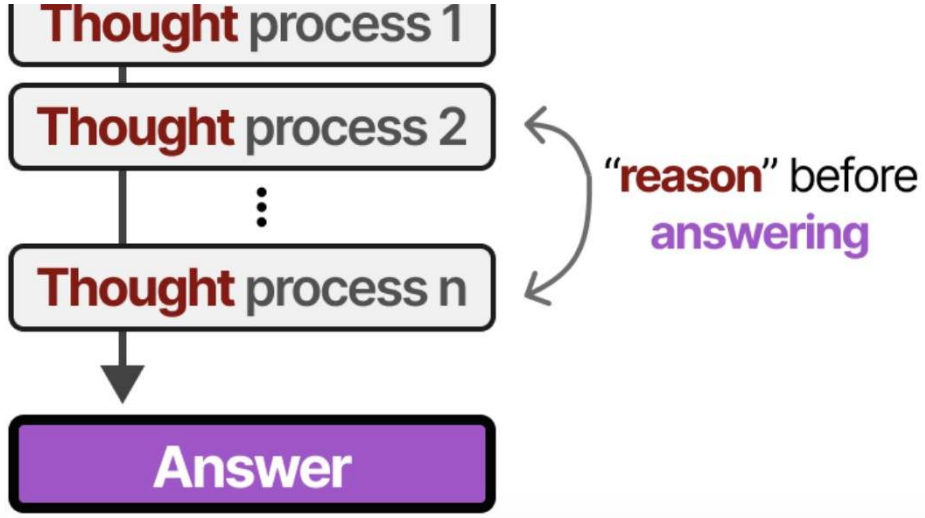
"Regular" LLMs

Question

"Reasoning" LLMs

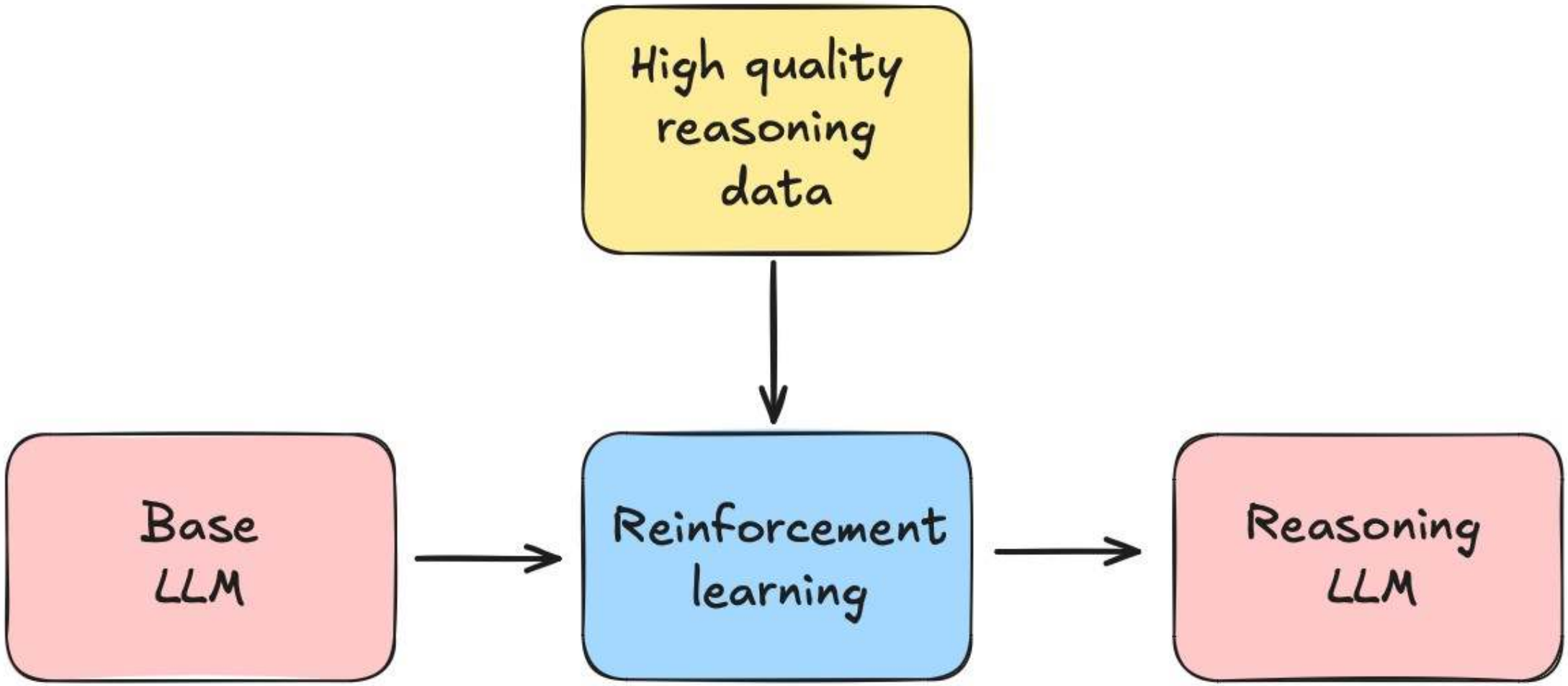
Question

- **Type 1 thinking** helps navigate everyday decisions rapidly, conserving mental resources but often at the cost of accuracy and susceptibility to errors.
- **Type 2 thinking** is essential for accurate, thoughtful decision-making, particularly in complex or novel situations, but consumes significant mental resources and time.



Daniel Kahneman in his book *Thinking, Fast and Slow*


Deep Seek used 600,000
reasoning examples to train R1



deepseek-ai/DeepSeek-R1-Distill-Llama-70B


On the fly reasoning approach
Chain of Thought + Tournament

Simple Test-Time Compute

~20x more expensive than single pass

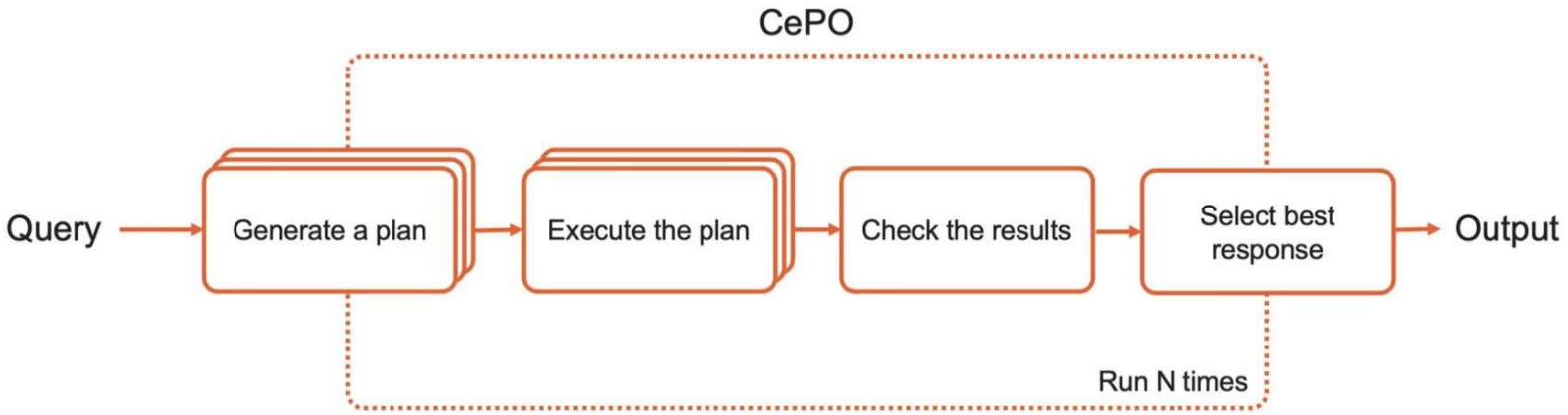
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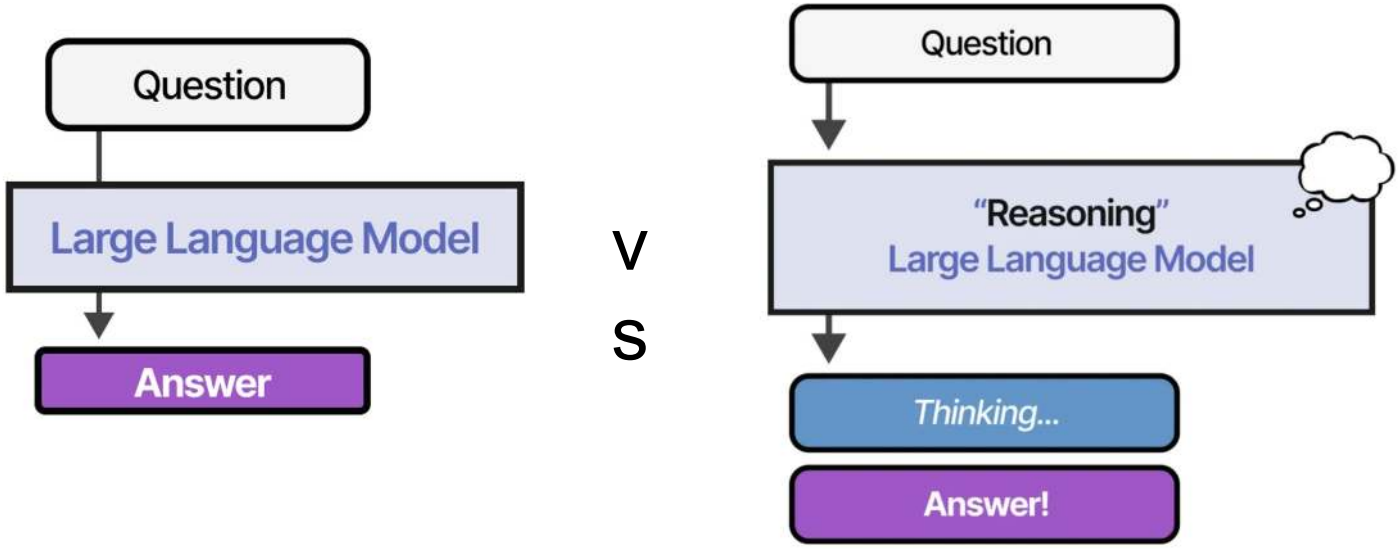
For FRG we did four runs on 200 multiple choice questions derived from research monographs

Scout — 56.5%
o3 — 87.5%
CePO (Scout) — 81.5% 25-point gain
o4-mini — 88.0%

For RBE we did a similar four runs on 200 multiple choice questions derived from research publications

Scout — 62.0%
o3 — 90.0%
CePO (Scout) — 84.0% 22-point gain
o4-mini — 91.5%






Resulting models are excellent coders!